Multidimensional Development of Pd-based Catalysts in Acetylene Hydrogenation: Effect of Gold and Carbon

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Plasma pyrolysis of hydrocarbons generates black carbon along with ethylene and acetylene[1]. Due to operational hazards, concentrated acetylene must be selectively hydrogenated into valuable ethylene. Pd-based catalysts are commonly used for this reaction. However, the surface and subsurface dynamics of active catalysts during acetylene hydrogenation are not yet fully understood. A rational design approach involves using 2-D Laterally Condensed Catalysts (LCCs)[2], which features a nonreactive functional interface between a thin metal layer (3 nm Pd, Pd-Au), a buffer layer (20 nm SiO_2), and a reactive interface exposed to the feed gas (**Fig. 1a**). These interfaces can be examined using operando spectro-microscopy. Pd-Au LCCs were employed to investigate catalyst multidimensional development during acetylene hydrogenation.

Au not only mimics atomic carbon in Pd LCC, affecting Pd local electronic properties but also temporarily introduces mesoscopic geometric effects by influencing the distribution of Pd atoms at the surface as well as the concentration of Pd:C and Pd:H species in the subsurface. In the absence of Au, more carbon diffuses into the Pd corroborated by the presence of a carbon interlayer beneath the spent Pd LCC (TEM). Long-time measurements (>260 hours) paradoxically showed that the superior initial selectivity (**Fig. 1b**) of Pd-Au LCC catalysts diminishes over time (**Fig. 1c**) due to Pd segregation to the surface (TEM). On the macroscopic scale, less surface carbon blockage was observed when adding Au (Raman). Additionally, Au in Pd-Au LCC mitigated catalyst agglomeration (SEM).

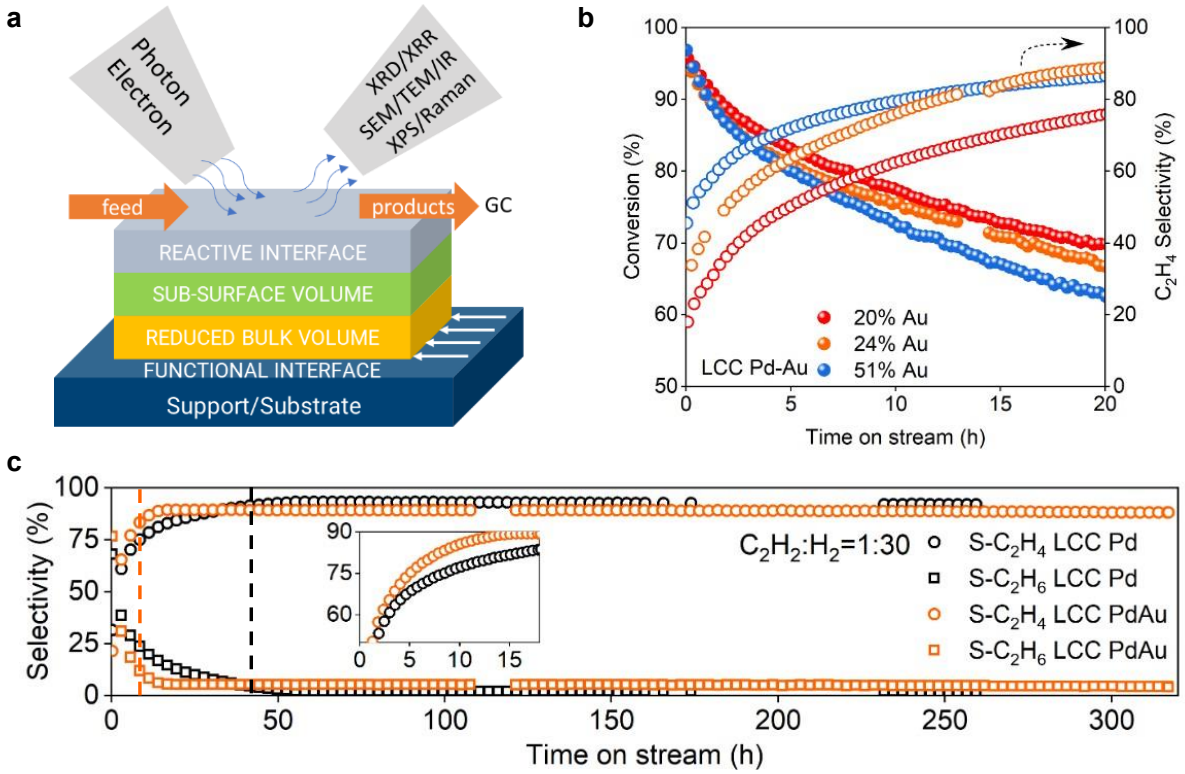

Figure 1. Laterally Condensed Catalysts Pd and Pd-Au. a, Schematic representation of the layer structured LCC catalyst including reactive interface, reduced subsurface volume, functional interface and support. **b**, Catalytic conversion and C_2H_4 selectivity of Pd-Au LCCs with Au (atomic) content of 20%, 24% and 51% in the metal film. Solid and open dots represent conversion and selectivity, respectively. **c**, selectivity (semi- and full-hydrogenation) of Pd and Pd-Au LCCs (TOS up to 315 hours). Inserts show details in formation phase before steady state (<18 hours). Reaction condition: C_2H_2: 0.9 ml/min; H_2: 27 ml/min; N_2: 5.8 ml/min; T = 150°C.

Reference

1 Gladisch, H. Acetylen - herstellung im elektrischen lichtbogen. *Chem. Ing. Tech.* **41**, 204-208 (1969). doi:10.1002/cite.330410416

2 Li, Z. et al. Rationally designed laterally-condensed-catalysts deliver robust activity and selectivity for ethylene production in acetylene hydrogenation. *Nat. Commun.* **in press** (2024)